

KPI identifier	G1.1
KPI name and unit	Primary energy [kWh/m2]
Equation	$\sum E_{Ptot} = \sum_i E_{ren,i} + \sum_i (E_{del,i} f_{del,tot,i}) - \sum_i (E_{exp,i} f_{exp,tot,i})$
Before/after ENTRANCE actions	
Details	Eren,i annual renewable energy produced on site or nearby for energy carrier i [kWh/m2] Edel,i annual delivered energy on site or nearby for energy carrier i [kWh/m2] f_{del,tot,i} total primary energy factor for the delivered energy carrier i, [-] Eexp,i annual exported energy on site or nearby for energy carrier I [kWh/m2 a] f_{exp,tot,i} total primary energy factor of the energy
Data required	
Available	Energy delivered to meters
Available from data analysis	On-site energy production
Not public	Energy exported to grid
Not available	
Available after action	
Estimation from data or literature	

G2.1	G2.2
Demand Response [%] $DR^p = \frac{p_h^{base} - p_h^{LS}}{p_h^{base}}$ Ph base baseline hourly power consumption [kWh] Ph LS hourly power consumption of a load using the DR profile [kWh] For each load (such as heat pumps, refrigerators, and air conditioners), two consumption profiles are calculated: the baseline setpoint profile, which is the load before DR solution and a DR setpoint profile which is the load after implementation of DR strategy (such as electricity price signal on the load profile or variable temperature set-point, etc.) in order to estimate the change	Peak Load reduction [MW] $\Delta P = P_{baseline, peak} - P_{flexible\ peak}$ calculate baseline peak as design capacity and compare to actual capacity
Energy delivered to meters	Energy delivered to meters
Energy delivered to meters after DR	Energy delivered to meters after DR

G2.4

Energy Shift Flexibility Factor

$$FS = \frac{\int_{lowpricetime} q_{heating} dt - \int_{highpricetime} q_{heating} dt}{\int_{lowpricetime} q_{heating} dt + \int_{highpricetime} q_{heating} dt}$$

Energy delivered to meters

Energy prices (electricity and heating)

G2.5

Flexibility Factor

$$FF = \frac{\int_{non\ peak} q_{heating} dt - \int_{peak} q_{heating} dt}{\int_{non\ peak} q_{heating} dt + \int_{peak} q_{heating} dt}$$

Energy delivered to meters

Peak demand periods

G2.11

Grid Interaction Index [%]

$$f_{grid,i} = \left[\frac{netgrid}{max | netgrid|} \right] \times 100$$

$f_{grid,i}$ grid interaction index [-]

netgrid: net grid metering over a given period (e.g. monthly)

compared to the maximum nominal contractual grid power given by contract with the energy company [kWh]

Power consumption (electric) delivered to meters

Maximal nominal contractual power

G2.12

No Grid Interaction Probability [-]

$$P_{E \sim 0} = \frac{\int_{\tau_1}^{\tau_2} dt |ne|(t) < 0.001}{\tau_2 - \tau_1}$$

PE~0 no grid interaction probability [%]

ne(t): normalized variable for the net exported energy [kWh]:

ne(t)= (export to grid-import from grid)/contractual power

τ1 and τ2: are the start and the end of the evaluation period

t: is the time

Energy delivered to meters

Energy exported to grid

Maximal nominal contractual power

G2.15

Energy Efficiency of Demand Response Action [%]

$$\eta_{ADR} = 1 - \frac{\int_0^{\infty} (Q_{ADR} - Q_{ref}) dt}{\int_0^{\text{length}} (Q_{ADR} - Q_{ref}) dt}$$

eta(ADR): %

Qadr: energy consumption during DR

Qref: baseline energy consumption

length: duration of the demand response event (denominator
calculates the total difference in energy consumption during DR

Energy delivered to meters

Duration of DR action

G3.1

Load Cover Factor [-]

$$y_{load} = \frac{\int_{\tau_1}^{\tau_2} \min[g(t) - S(t) - \zeta(t), l(t)]dt}{\int_{\tau_1}^{\tau_2} l(t)dt}$$

- S(t)= Sc-Sdc
- yload load cover factor [-]
- g(t) on-site generation [kWh]
- S(t) storage energy balance [kWh]
- Sc charging storage energy [kWh]
- Sdc discharging storage energy [kWh]
- ζ(t) storage energy losses [kWh]
- l(t) building load [kWh]
- t time
- τ1 and τ2 are the start and the end of the evaluation period

Energy delivered to meters

On-site energy production

Storage energy balance

Storage energy losses

Discharging storage energy

G3.2

RES Self-consumption (Supply Cover Factor) [%]

$$\phi_{SC} = \frac{\int_{t=1}^{t2} M(t)dt}{\int_{t=1}^{t2} P(t)dt}$$

ϕ_{SC} Self-consumption [-]
M(t) instantaneously overlapping of the generation and load profiles [kWh] M(t)=min{L(t),P(t)}.
L(t) instantaneous building power consumption [kWh]
P(t) instantaneous on-site RES power generation [kWh]

Energy delivered to meters

On-site energy production

G3.7

Load Matching Index [%]

$$f_{load,i} = \frac{\sum_{T1}^{T2} \min[1, \frac{g(t) - S(t) - \zeta(t)}{l(t)}}{N}$$

$g(t)$ local renewable energy generation at time step
 $S(t)$ storage energy balance [kWh]
 $\zeta(t)$ energy losses (sum of generation energy losses, storage energy losses, building technical systems energy losses (excluding storage) and load energy losses (e.g.: distribution losses)) [kWh]
 $l(t)$ building load [kWh]
 N is the number of samples in the evaluation period, from τ_1 to τ_2 .
In case that hourly resolution data are used, and the evaluation period is a complete year, the number of samples is 8760.

Energy delivered to meters

On-site energy production

Storage energy balance

Storage energy losses

Discharging storage energy

Data granularity

G6.1

System Average Interruption Duration Index (SAIDI) [failures/year and customer]

$$SAIDI = \frac{\sum r_i N_i}{N_T}$$

SAIDI System Average Interruption Duration Index [hours/year and customer]

N_i is the number of customers [-]

r_i total average down time [h/year]

N_T total number of customers served [-]

Number of nodes

Number of nodes affected by interruption

Duration of system interruption time (electricity or heating)

G6.2

System Average Interruption Frequency Index (SAIFI)

$$SAIFI = \frac{\sum \lambda_i N_i}{N_i}$$

SAIFI System Average Interruption Frequency Index

λ_i total average failure frequency [failures/year]

N_i total number of customers for i [-]

Number of nodes

Number of nodes affected by interruption

System failure frequency

G7.1

Thermal comfort: predicted mean vote

$$PMV = [0.303 \exp(-0.036M) + 0.028]L$$

Metabolic rate

L: thermal load is defined as the difference between the internal heat production and the heat loss to the actual environment for a person hypothetically kept at comfort values of skin temperature and evaporative heat loss by sweating at the actual activity level

W the rate of mechanical work accomplished (W/m²);

t_a the air temperature;

t_r the mean radiant temperature;

f_{cl} the clothing area factor, dimensionless;

Metabolic rate of occupants based on room final use

Ambient temperature

Clothing factor

convective heat transfer coefficient

mean radiant temperature

mean clothing temperature

water vapor pressure in ambient air

clothing thermal resistance

$$L = (M - W) - 3.96 \times 10^{-8} f_{cl} [(t_{cl} + 273)^4$$

$$\begin{aligned}
& - (t_r + 273)^4] - f_{\text{cl}} h_c (t_{\text{cl}} - t_a) \\
& - 3.05[5.73 - 0.007(M - W) - p_a] \\
& - 0.42[(M - W) - 58.15] \\
& - 0.0173M(5.87 - p_a) - 0.0014M(34 - t_a),
\end{aligned}$$

$$\begin{aligned}
t_{\text{cl}} = & 35.7 - 0.028(M - W) \\
& - R_{\text{cl}}\{39.6 \times 10^{-9} f_{\text{cl}}[(t_{\text{cl}} + 273)^4 - (t_r + 273)^4] \\
& + f_{\text{cl}} h_c (t_{\text{cl}} - t_a)\},
\end{aligned}$$

G7.2

Thermal comfort: predicted percentage of dissatisfied

$$PPD = 100 - 95 \exp[-0.03353 PMV^4 + 0.2179 PMV^2].$$

PPD represents the percentage of people likely to be dissatisfied with the thermal environment:

Predicted mean vote

G7.3

Indoor overheating degree

$$IOD = \frac{\sum_{z=1}^Z \sum_{i=1}^{N_{occ}(z)} [(T_{op,i,z} - TL_{comf,i,z})^+ \bullet t_{i,z}]}{\sum_{z=1}^Z \sum_{i=1}^{N_{occ}(z)} t_{i,z}}$$

Z [-] is the total number of building zones with occupancy

z is zone counter,

Nocc(z) [-] is the total number of occupied hours in zone z,

i is the hour counter,

Top,i,z [°C] is the indoor operative temperature in zone z at hour i,

TLcomf, i,z [°C] is the comfort temperature limits in zone z at hour i.

Number of zone supplied by HVAC system

Occupied hours

Operative temperature

Comfort temperature

G7.4

Operative temperature [°C]

$$T_o = \frac{(h_r t_{mr} + h_c t_a)}{h_r + h_c}$$

Room temperature

Room design

G7.6

CO2 concentration [ppm]

$$C = \frac{q_{CO2} * 1000}{q_{sp} * 3.6} \quad q_{sp} = \frac{q_s}{n}$$

qCO2: generation rate [L/h],
qsp: ventilation rate per person,
qs: ventiaction rate,
n: number of people

Air flow

Occupants emisssions based on room final use

number of people (estimates)

G7.7

Indoor Climate Class (IDCC)

$$c^* = \frac{m}{L} \cdot 10^6$$

c^* is the equilibrium CO₂ concentration
above ambient level, ppm;
 m is emissions from occupants, $\text{l}(\text{s} \cdot \text{m}^2)$;
 L is total air flow, $\text{l}(\text{s} \cdot \text{m}^2)$.

Occupancy (static or dynamic)

G7.8

Unified Glare Rating (UGR) [UGR]

$$UGR = 8 \log \left[\left(\frac{0,25}{L_b} \right) \sum_i \frac{L_i^2}{P_i^2} \right]$$

L_b is the background luminance (cd/m²),
 L_i is the luminance (cd/m²) of each luminaire,
 ω_i the solid angle [sr] in the direction of the observer,
 P_i the position index of each luminaire.

G8.1

Investment cost [€]

CAPEX

Technology implementation costs

G8.2

Specific cost of flexibility [c€/kWh]

$$C_{sp} = \Delta C / |\Delta E|$$

The ratio between the extra cost due to the load shifting ΔC [c€] and the amount of electricity ΔE [kWh] shifted out compared to the reference

Extra cost that follows from the load shifting ΔC

Amount of energy shifted from peak

G8.3

Flexibility Spread (flexibility saving index) [€/kWh]

$$FSI = \frac{\text{Cost of flexible operations}}{\text{Cost of baseline operations}}$$

express the convenience of flexible solutions compared to the reference scenario

Costs before action

Cost after actions